Lesson plan · for the teacher

# One ID per pupil: joining sheets

Data engineering · Level 2 · Lesson 4

**Time**
40 min

**Materials**
Class-list cards (a number and a made-up name) and mark cards (a number and a mark); Board and chalk; Optional: a spreadsheet app with XLOOKUP or VLOOKUP, and Python

## Learning goals

- Give every pupil one ID.
- Join two sheets by ID, not by name.
- Find and question the rows with no match.

## Activity: Human join

1. Warm-up (5 min): Show a short mark list with two pupils of the same name and one spelling difference. Ask pupils to match the marks to the class list by name, and see where it goes wrong.
2. Teach (8 min): Explain a key, a pupil ID given once and never reused, and how a lookup finds the matching row and brings back the name.
3. Join (12 min): Hand out the class-list and mark cards. Pupils find their partner by number and stand together; unmatched pupils stand to one side.
4. Question (10 min): List the unmatched cards and the pupil with two marks on the board. For each, the class decides what it might mean and who to ask.
5. Wrap-up (5 min): If you have a device, show the same join with XLOOKUP or a Python dictionary. Then give the exit ticket.

**Prepare the cards before the lesson: one pupil with no mark, one mark with an unknown number and one pupil with two marks. Use made-up names, and keep the number cards so you can reuse them.**

## Discussion

- Why do names make poor keys?
- What could a mark with no pupil mean?
- Why did one pupil appear twice?

## Exit ticket

- Write what a key is in one sentence, and one reason a row might have no match.

**Try it at school tomorrow**
Play “Human join”: half the class holds class-list cards (a number and a name), the other half holds mark cards (a number). Pupils find their partner by number. Who’s left without a partner, and why?

Practice questions · for you

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**1** **After a lookup, one mark has no matching pupil. What does that tell you?**

- ◯ The lookup formula is broken
- ◯ Delete it: it's rubbish
- ◯ An ID was typed wrong, or a pupil is missing from the list

**2** **Two pupils may share the same pupil ID if they are in different classes.**

- ◯ True
- ◯ False

**3** **What does this program print?**

```
names = {1: "Amina", 2: "Tunde", 3: "Grace"}
print(names.get(2, "?"), names.get(7, "?"))
```

- ◯ Tunde ?
- ◯ Tunde Grace
- ◯ An error: there is no pupil 7

**4** **pupils.csv is the class list and test.csv holds the test marks by pupil ID. Fix the program so it prints each mark with its pupil's name, and ? when the ID isn't on the list, instead of stopping with an error.**

```
import csv
names = {}
with open("pupils.csv") as f:
    for row in csv.DictReader(f):
        names[row["id"]] = row["name"]
with open("test.csv") as f:
    for row in csv.DictReader(f):
        print(names[row["pupil_id"]], row["mark"])
```

Write your program:

**5** **Put the lines in order (and indent them): the program prints every pupil on the class list who has no mark.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

- ☐ `marked.add(pupil_id)`
- ☐ `if pupil_id in marked:`
- ☐ `for pupil_id, mark in marks:`
- ☐ `if pupil_id not in marked:`
- ☐ `marked = set()`
- ☐ `for pupil_id in pupils:`
- ☐ `print(pupil_id, "has no mark")`

✎ **Does your school give every pupil one ID that's used on every sheet? If not, which two sheets would you join first, and what would you learn?**

Check your answers

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**1** **An ID was typed wrong, or a pupil is missing from the list**

A mark with no pupil points to a typing mistake in the ID, or a gap in the class list.

**2** **False**

An ID is a key only if it picks out exactly one pupil in the whole school.

**3** **Tunde ?**

names.get(2, "?") finds pupil 2, Tunde. Pupil 7 isn't in the dictionary, so .get gives the "?" instead of stopping with an error, as names[7] would.

**4**

```
import csv
names = {}
with open("pupils.csv") as f:
    for row in csv.DictReader(f):
        names[row["id"]] = row["name"]
with open("test.csv") as f:
    for row in csv.DictReader(f):
        print(names.get(row["pupil_id"], "?"), row["mark"])
```

The dictionary holds each name under its ID, so every mark finds its pupil in one step.

**5** **Top to bottom: 3, X, 2, 5, 1, 4, 6**

```
marked = set()
for pupil_id, mark in marks:
    marked.add(pupil_id)
for pupil_id in pupils:
    if pupil_id not in marked:
        print(pupil_id, "has no mark")
```

First collect every ID that has a mark in a set.